

Data Mining Questions

Clustering / K-Means Clustering

1. What is clustering? Compare among different Hierarchical clustering similarity criteria.
2. Suppose that the data mining task is to cluster points (with (x, y) represent location) into three clusters, where points are $A1(2, 10)$, $A2(2, 5)$, $A3(8, 4)$, $B1(5, 8)$, $B2(7, 5)$, $B3(6, 4)$, $C1(1, 2)$, $C2(4, 9)$. The distance function is the Euclidean distance / Manhattan distance / Minkowski distance. Suppose initially we assign $A1$, $B1$, and $C1$ as the center of each cluster, respectively. Use the k-means algorithm to show only
 - a. The three cluster centers after the first round of execution
 - b. The final three clusters
3. What are the challenges when clustering is applied to a real-world dataset?
4. Given two objects represented by the tuples (22, 1, 42, 10) and (20, 0, 36, 8)
 - a. Compute the Manhattan distance between two objects
 - b. Compute the Minkowski distance between the two objects, using $q=4$.
5. Write short note on K-means algorithm.

Classification / SVM

1. Mention the roles of Support Vector Machine (SVM)
2. Define kernel trick. Discuss about different kernel Support Vector Machines.
3. How is non-separability of data handled in SVM learning?
4. Describe a Support Vector Machine. Define the optimization task solved in SVM learning with an example.
5. In Support Vector Machines, what is a support vector? What is meant by the margin to be maximized?

Dimensionality Reduction / PCA

1. What is understood by curse of dimensionality? How can we avoid this problem?
2. What is data reduction? Explain dimensionality reduction for feature extraction and feature selection processes.
3. Briefly discuss the steps of Principle Component Analysis (PCA) technique
4. How dimensionality reduction helps in data mining
5. What are the necessities of dimensionality reduction

6. Justify the use of PCA in various applications like noise reduction, feature extraction, feature reduction etc.

Association rule mining / Apriori Algorithm

1. Explain the generation of frequent item sets using Apriori algorithm using suitable example.
2. What are association rule in data mining. Give the names of algorithm in association rule.
3. What are use cases of association rule in data mining?
4. How do association rules work?
5. Write short notes on Apriori algorithm.
6. What are the components of Apriori algorithm?
7. From the table below, find frequent item-sets and generate the association rule using the Apriori algorithm: (*maximum support = 2, minimum confidence = 50%*)

TID	ITEM SETS
T1	A, B
T2	B, D
T3	B, C
T4	A, B, D
T5	A, C
T6	B, C
T7	A, C
T8	A, B, C, E
T9	A, B, C

Introduction / Common

1. In context of machine learning, define training data, testing data, data mining, supervised, unsupervised and reinforcement learning
2. Define feature, feature extraction and data processing with example
3. What are the primary differences between machine learning, data mining, and pattern recognition?
4. From the following confusion matrix, find out –

N = 165	Predicted: NO	Predicted: YES
Actual: NO	50	10
Actual: YES	5	100

Accuracy, Precision, Recall, F1 – score, FN rate, TN rate

5. What are outliers? How might you determine outliers in the data?
6. What is the basic difference between supervised and unsupervised learning?
7. What is data mining? In your answer, address the following:
 - a. Is it another hype?
 - b. Is it a simple transformation of technology developed from databases, statistics, and machine learning?
 - c. Explain how the evolution of database technology led to data mining.
 - d. Describe the steps involved in data mining when viewed as a process of knowledge discovery
8. Suppose your task as a software engineer at Big University is to design a data mining system to examine the university course database, which contains the following information: the name, address, and status (e.g., undergraduate or graduate) of each student, the courses taken, and the cumulative grade point average (GPA). Describe the architecture you would choose. What is the purpose of each component of this architecture?
9. Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.
 - a. What is the mean of the data? What is the median?
 - b. What is the mode of the data? Comment on the data's modality
 - c. What is the midrange of the data?
10. Why accuracy isn't the most suitable metric for the classification of the imbalanced dataset?
11. Explain the importance of Data Cleaning in Data mining? Describe the major tasks in Data cleaning.
12. How to handle missing values in real world data. Describe various methods.